Divide and Conquer Algorithm Efficiency Evaluation
Analyze a divide-and-conquer algorithm with recurrence $T(n) = 2T(n/2) + n \log n$.

CSA0613-Design and Analysis of Algorithms

CO1-ASSESSMENT TOOL 03-TECHNICAL PRESENTATION

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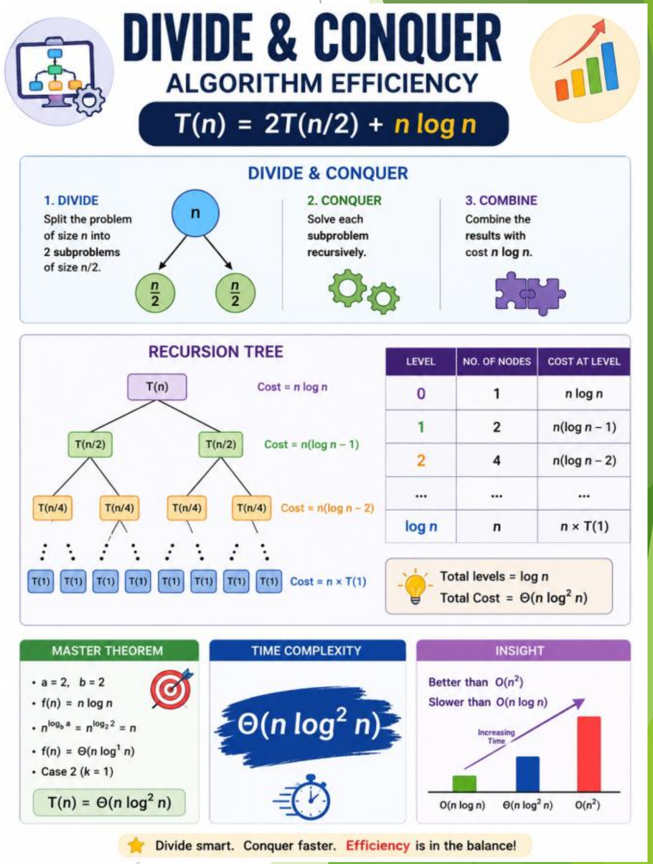
INTRODUCTION

- Divide and Conquer is an algorithmic technique that solves a problem by dividing it into smaller subproblems of the same type, solving them recursively, and combining their results to obtain the final solution. This approach improves efficiency by reducing the complexity of large problems and is used in algorithms such as Merge Sort, Quick Sort, and Binary Search.

ENGINEER TO EXCEL

PROBLEM STATEMENT

- Analyze the divide-and-conquer algorithm with the recurrence (T(n)=2T(n/2)+n\log n).
- Explain how the problem is divided into two equal subproblems and solved recursively.
- Apply the **Master Theorem** to determine the algorithm's time complexity.
- Illustrate the recursive breakdown using a recursion tree.
- Interpret the result and justify the efficiency of the algorithm for large input sizes.



ALGORITHMS & PSEUDOCODE

Algorithm

1.Start.

2.If $n = 1$, return the base case value.

3.Divide the problem into two equal subproblems of size $n/2$.

4.Recursively solve both subproblems.

5.Combine the two solutions with a cost of $n \log n$.

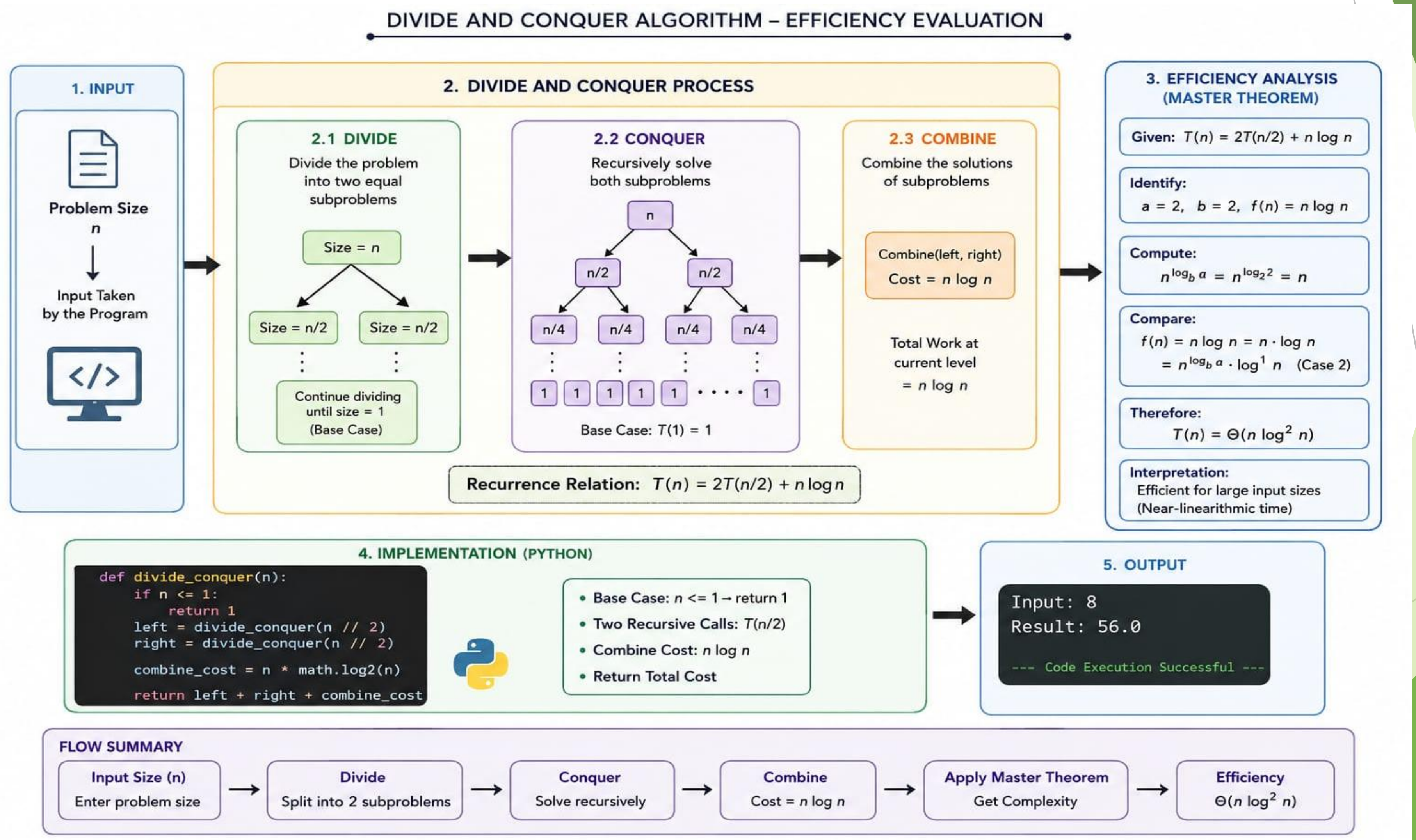
6.Return the combined result.

7.Stop.

Pseudocode

```
Algorithm DivideConquer(n)

if n == 1 then
   return BaseCase

left  = DivideConquer(n/2)
right = DivideConquer(n/2)

Combine(left, right)    // Cost = n log n

return Result
```

ARCHITECTURAL DIAGRAM



IMPLEMENTATION

```python
import math
def divide_conquer(n):
    if n <= 1:
        return 1
    left = divide_conquer(n // 2)
    right = divide_conquer(n // 2)

    combine_cost = n * math.log2(n)

    return left + right + combine_cost
n = 8
result = divide_conquer(n)
print("Input:", n)
print("Result:", result)
```

OUTPUT

```
Input: 8
Result: 56.0

=== Code Execution Successful ===
```

CONCLUSION

- The divide-and-conquer algorithm divides the problem into two equal subproblems and solves them recursively.
- Applying the **Master Theorem** gives the time complexity as $\boldsymbol{\Theta(n \log^2 n)}$.
- The recursion tree shows that there are $\boldsymbol{\log n}$ levels of recursive calls.
- The algorithm is more efficient than $\boldsymbol{O(n^2)}$ algorithms but slower than $\boldsymbol{O(n \log n)}$ algorithms.
- Overall, the algorithm is suitable for handling large input sizes with balanced recursive division and efficient performance.

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THANK YOU